

VARIANCE & STANDARD DEVIATION

→ $\mu = E(\bar{X})$ - measures "center of mass" of distribution

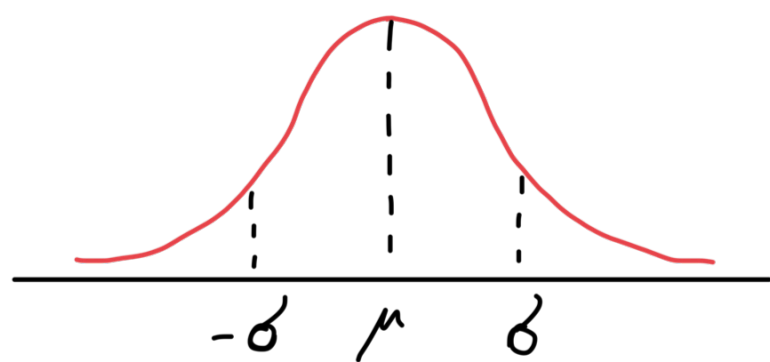
→ $\text{VAR}(\bar{X})$ - measures avg. square deviation of $\bar{X}$ from μ

$\text{SD}(\bar{X})$ - measures the spread of histogram

$$\text{SD} = \sqrt{\text{VAR}(\bar{X})}$$

$$\text{VAR}(\bar{X}) \triangleq E((\bar{X} - \mu)^2)$$

$$\stackrel{d}{=} E(X^2) - [E(X)]^2$$



$$\text{VAR}(\bar{X}) = E((\bar{X} - \mu)^2) = E(\bar{X}^2 - 2\mu\bar{X} + \mu^2) =$$

$$= E(\bar{X}^2) + E(\underbrace{-2\mu\bar{X}}_{-2\mu^2}) + E(\underbrace{\mu^2}_{\mu^2}) =$$

$$= E(\bar{X}^2) - \mu^2$$

$$E(\bar{X}^2) = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$\bar{X} \sim N(\mu, \sigma^2) \Rightarrow f(x) = \frac{1}{\sqrt{2\pi} \sigma} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

$$\Downarrow \\ E(\bar{X}) = \mu$$

$$\text{VAR}(\bar{X}) = \sigma^2$$